

BRAC University, Dhaka
Department of Computer Science and Engineering

CSE 446: Blockchain and Cryptocurrencies, Spring 2026, Quiz - 2

Marks: 20

Time: 20 Min

Instructions:

- i) Add your name and registration number on the top of your answer script.**
- ii) When submitting the scanned pdf answer script, please rename the file as yourID_Quiz2_CSE722**
- iii) Upload your pdf answer script using this Google form: <https://forms.gle/LvwmeNFEkBweTxxe8>**

Name:	ID:	Section:
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1. Determine if the following sentences are true or false. For any false sentence, write its correct form. 4*1 = 4
 - i. Every Txout is essentially a Txin. (F)
Every Txin is actually a Txout of a previous tx.
 - ii. Bitcoin uses an account based ledger. (F)
Bitcoin uses a transaction based ledger.
 - iii. A private key is hashed to create a Bitcoin address. (F)
A public key is hashed to create a Bitcoin address.
 - iv. Every blockchain is a distributed ledger. (T)
2. Fill in the blanks 4*2 = 8
 - i. Two advantages of Bitcoin over Fiat money are any of these two: easily portable, deflationary and need to prove ownership
 - ii. A transaction output in Bitcoin has two fields such as: value & scriptPubKey
 - iii. A custodial wallet is dangerous because the private key of user will not be under his/her control
 - iv. UTXO set shrinks when the number of txouts is smaller than the number of txins
3. Briefly answer the following questions. 4*2 = 8
 - i. Differentiate between Hybrid and Consortium Blockchain.
Read access is open and write access is restricted in a hybrid blockchain where both read and write accesses are restricted.
 - ii. Why is scriptPubKey considered a locking script?
Because scriptPubKey locks the particular amount for the address specified in the script. Only the owner of that address can unlock that amount using scriptSig.
 - iii. State two advantages of using public key schemes for Decentralized Identity.
i) Users can create as many accounts as required and ii) nobody can confiscate or shutdown decentralized accounts.
 - iv. Determine the fee of Tx1 assuming Tx0[0] is 3.125BTC.

Tx1	
Tx0[0]	Amount: 1.50 To: Leonard
	Amount: 1.45 To: Raj
	Amount: 0.17 To: Sheldon

Fee = 3.125 – (1.5 + 1.45 + 0.17) = 0.005